

ActInf Textbook Group ~ Cohort 5 ~ Session 7 (Chapter 3, Part 2) ~ 10/31/23

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 7 (Chapter 3, Part 2) ~ 10/31/2023 | 56:34

all right greetings everyone welcome

back it's Halloween 23 and we're in our

second discussion on chapter 3 so we

will look at the cohort 5 questions on

chapter

3 um

so I encourage people to add even now

questions directly to cohort 5 questions

um or the or or or there's all this

other questions so maybe people added

directly to the table and so it doesn't

have the cohort five tag but before we

go to any questions does anyone just

want

to bring up some piece of chapter 3 that

stuck with them or highlight some

other topic or share a little bit about

how they're seeing the low road and the

high road now that we're getting to our

second

discussion on the high

road yes Susan then anyone else who

raises their

hand so does the low road and the high

road are they

differentiated by active inference

versus Active

Learning no Active Learning was the term

used by that one paper just to sort of

spin the fast ball a little bit

different to highlight kind of like a

regime that an active inference agent

could be in instead of a purely confir

confirmatory sensory mode rather a mode

that that reduces uncertainty maximally

about world States but active inference
is the bigger Circle which what the
authors characterized as active learning
is a smaller subset Within
which incorporates it um and um so
Dean um brought up that active
inference was a process of
provisioning can you speak to
that we would have this was in private
conversation so we'd have to know a
little more about what he meant um but
but it's a very
evocative yeah so that but that that
kind of brought the question of
differentiating active learning from
active
inference versus belief updating so
because learning would involve belief
updating
right yeah as as with a huge number of
the topics that we hear about belief
updating attention learning
anticipation there's often a narrower
technical sense like belief updating is
just any basian parameter updating at
all so belief updating doesn't always
refer to a human with a psyche
undergoing some kind of like narrative
development it can be a lot more narrow
and small and less loaded with like
philosophical or cultural baggage than
if we were to say Hey when's the last
time you had a belief updating most
people aren't going to say Well when one
of my proprioceptors in my arm you know
did something different
um so sometimes yeah the narrow sense or
the technical sense of a

term and then the the broader
um um scope of what it might
evoke and and so these are all good
comments um Kristoff then anyone else
who raises their
hand uh I I just want to say that the
high road and and sort of
the kind of um uh you know the road to
active inance through this ideas of
self-organization and self- evidencing
that that really reminded me of um some
work that I'm not familiar with uh
deeply mathematically but conceptually
seems very similar that's um German
England's work on the thermodynamic
origin of life he kind of um in his work
he has shown that if you have um uh
system that's receiving an energy from
from external source and it has a place
to dissipate that energy and there are
certain other conditions that need to be
met then the system will necessarily
organize itself in such a way uh to take
advantage of that energy source um so it
will start creating structures that will
allow it to um um to uh you know utilize
that energy and in his talk I think that
was at MIT a couple years ago you know
at the very end of the talk very
kind of he makes this offhand remark
that probably um when the energy coming
in has some kind of temporal structure
that may eventually lead to the
emergence of intelligence or something
like that that kind of made me when when
when he said that in talk made me think
about you know faradise speculating
about the electromagnetic nature of

light uh he's he's probably right and
I'm pretty sure that he's I'm quite sure
because he started with thermodynamics
with his work that it can be recog
consed um with active inference so it's
also kind of fascinating that there is
this very close but slightly different
view that kind of arise arrives at
pretty much the same um same kind of
conclusion from you know um slightly
quite different set of assumptions but
related yeah very interesting Dr
Einstein than anyone
else um

this isn't low road High Road specific
so if you want to shelf it for a second
but I was curious about something on
page 52 that goes back to the
conversation we were having
here um two weeks ago in the in the
morning group or wherever you guys are
at and it's about the idea of perception
is

action um I remember we were getting
into that and on 52 you know they um use
this example

of if you're hungry and your your
glucose is low so you have a hunger
response and so you know there's this
line that says

um perception cannot go further than to
reduce your

surprise and it got me thinking about
like if anybody here is ever

fasted for for example you know you'll
be really hungry the first day or
whatever and then

by the second day your hunger is just

dissipated it

and that that got me thinking about that conversation we were having here regarding perception is action because you know what is it in terms of this field I mean obviously that's like a belief updating about your need to eat or something but

what what would be the mechanism you know uh according to this theory that would cause your hunger to dissipate because there's been no act there's been no action in the external world to reduce your hunger it seems like it would be purely just a matter of perception and so when when they have that one line that perception you know can go no further than to reduce the surprise

I yeah it just made me think about that kind of as a thought

experiment cool yeah I know a little bit about that this is andw okay go for it go for it yeah there there is action going on your body's metabolism is changing and um there is a yeah there's there's essentially external states that are functioning on your mind um in that transition between day one and day two on fasting that makes it not a completely internal State derived activity I mean would that count as taking an action though I mean because you're not consciously you know changing your external environment I think that if you sketch it according to the active ontology it'll bring it into the rails if you sketch

out what's being observed what are the
hidden States being inferred it will
clarify a lot of pie because otherwise I
mean we could go infinitely into the
actual

biomolecular processes that happen in
and fasting but to kind of return that's
not my question yeah yeah but that's
everything is how how do we take that
situation of Interest with its
particularities and approach it from The
High

Road Francis and then and then Kristoff
so uh from counseling background I'd say
that you have to look at the whole self
the part of you that needs the need for
food is one of your
needs you have other needs maybe the
need

to uh fit in with your uh Group whatever
need is driving you to
fast um so you have to look
at uh the perceptions goals of the whole
self and how these balance
out uh you're always dealing with
conflict in a conflict we have multiple
needs at any one

time um so

uh uh yeah um and how that would work at
the Active inference level I'm not in
entirely sure maybe it would uh
reinforce the

uh ex uh strengthen uh the expectations
in such a way that uh you ignore the um
evidence

against uh I don't I don't have a
sufficiently good strong understanding
to uh propose the

mechanism that's what I wanted to say
it's the whole

self cool Mark Miller has explored this
kind of like multi-drive balancing and
about well-being as optimizing well
across multiple domains because you
can't just like optimize your blood
glucose and not not your blood oxygen
you die um and then also internal or
covert action as attention to Simply
disend or to

attenuate from the grumbling of the
stomach helps make it an
unsurprising path of policy selection
not to order something at the restaurant
or put the fork in the mouth or whatever
it may be um Andrew then

Kristoff sure uh not just a general
thought of um regarding the active
inference framework I mean could we say
something like um you know someone is
making a decision to fast is a sort of
like a higher order decision but
nonetheless on the day today if they're
not used to fasting or even if they have
in the past but you know have not been
for say the past week such that they are
very used to and therefore expecting to
eat at regular timing intervals on the
day today that despite the fact that
they've made this kind of higher order
decision to not eat uh nonetheless like
they their body uh and other processes
are expecting food um they are in fact
not receiving food because of the
fasting and so there's a higher level of
something like there's an increase of
something like surprisal or or

uncertainty that is like even though I
can kind of consciously think like oh no
I could eat food I'm just choosing not
to my body is still experiencing
physiological shifts and changes because
I'm not eating that is generating a lot
of like um kind of uncertainty in my
body you know this is different this is
not usual this is not realizing my
typical set of preferences um and then
over time as the body kind of adjusts to
that um suddenly the need you know
hunger other um kind of sensations of
surprise or otherwise might start to
reduce is you the body starts to realize
like oh no I'm fine like my my
homeostasis is not exactly in danger
here even though you know it was kind of
a shock initially to start experiencing
the um you know the the the difficulties
of of fasting uh in the first place on
day one right so yeah here we have the
glucose observations and then you can
infer what the hidden state is again
it's the modeler's choice map not
territory but this could be risk of
death so it might be a matrix learning
oh actually I don't have a high risk of
dying even when it's at this value or
not then you mentioned like habits I
mean oh it's 5:00 pm so the generative
models in the business of the path of
least action doing the likeliest thing
so if 99% of the last thousand to 5 PMS
you did something your prior's high on
simply doing it as a matter of habit
or of having a preference for high
glucose would then lead to the selection

of policies that make those observations
unsurprising so once you start to kind
of take a figure or some
formalism and go beyond just associating
components of the territory to bringing
them in to stitching together a total
generative model you find that there's
there's many generative models you can
draw sometimes with convergent insights
other times that highlight different
features and that's part of the iterated
process of modeling is like developing a
portfolio of generative models like many
maps of the city and then finding out
well where are these special cases or
generalizations of each other what's the
kind of measurement or
experiment that as a modeler is going to
help me
differentiate amongst these two
different structural
hypotheses
Kristoff then anyone
else um you kind of meet me to the punch
because I was about to say that it's
it's all about tuning well it's not all
about but it's largely about tuning
Precision parameters um so if you if you
do an experiment where you fast
regularly like say every week you're
going to find it that you know even
though you're you're starting from
effectively the same starting point so
it's it's a somewhat reproducible
experiment week after week after week
after week it's becoming easier and
easier and easier to fast
and and also you know like I'm not I'm

not an adapt meditator but I have some level of skill and in my case um I can easily distinguish between two kinds of hunger there's a intellectual craving for novelty which I I derive quite a lot of novelty from from food it's a it's a pretty easy source of novelty and a lot of people I think um uh you know like they feel this uh they're feeling of hunger is kind of virtual that way um and there is a physiological hunger that's that's a very distinct and different kind of feeling um and most of the time I actually know that my body does not need food even though I I feel like I would like to eat something because I'm creating this um this intellectual novelty and I think there's an interesting way to actually experience this very directly like how this process of tuning Precision parameters um happens is that if you take you know a big exhale and empty your lungs and hold your hold your breath for as long as as you can at some point your body will for for you to take a breath you can't really just um you know suffocate that way because a mdala will kick in and they will trigger you know this kind of panic fight ORF flight response that will kick you into taking a big breath but if you do it again and again and again very quickly you just get used to it and you're no longer reacting to to that with panic and you can uh hold your breath for much longer and so on so forth and you can actually experience some really nice

effects of um saturating your blood with CO₂ which which is which is extremely relaxing um and it pushes like O₂ deeper into the the cells and actually helps to pry oen um away from you know um red blood cells and so on so it's and it's it's entirely controlled by amygdala and that's why it comes with this like a fascinating reaction and you can just tune your you can observe yourself tuning that that parameter and very quickly you can get used to much higher concentrations of CO₂ in your in your blood and it's not a problem at all okay cool just to sort of look ahead to the kinds of generative models that makeup doing active inference modeling we see this all the time figure 4.3 we're going to revisit chapter 7 so this is the partially observable Markov decision process now as shown here there are no Precision parameters but then quickly after they show well yeah you can actually have another parameter kind of on top of each of them little D on top of the D prior little a on top of the a little B on top of the B and so on and this speaks to the extensibility or the modularity of building generative models using base graphs is in the kind of plain first level model you just have a fixed a Matrix and then you can say well what about having a variance on the a matrix like an like a kind of precision on a well if that helps you get at the phenomena that you're interested in that might be something to try you could also

continue to even develop further say
well what if we had policies on which
attention we wanted to have on
a no one's stopping you it's just map
building and especially if it's um being
done as a kind of thought
experiment outside
of
reconnecting with the empirical
measurements then yeah there's basically
no end to
speculating about where you could it's
just like building
Lego
constructions because you could you
could have another variable connected to
here and here and here and here and so
that's hence it can be kind of a
open-ended and aesthetic journey to make
these kinds of glyphs but then if you
want to reconnect it back to the
empirical data which is discussed most
in chapter nine then you kind of have a
Divergent convergence iterative process
where you expand the portfolio of
generative models you have and then you
use some experiments or measurements to
then select or do basian model selection
to go back down to a smaller subset of
them and then you can kind of diversify
from there select them back
down
um any other thoughts or we can look at
some of the most um uploaded questions
or any other questions that people want
to look
to
any any other thoughts or or just

remarks on the low road and the high road um I I did uh just kind of want to throw it out there like near the end of the high road like box 3.3 they're describing entropy minimization and open-ended behavior and I guess just I just want to mention generally the topic of like how does an agent within a fluctuating environment uh on the one hand uh find means of like adapting to that environment like that is a h how does it act in order to maintain something like homeostasis or allostasis but at the same time not become too rigid like not repeatedly make the same exact decisions purely to maintain its its homeostasis and instead uh allow for something like improvisation or um like they phrased it as an open-ended repertoire of Novel behaviors creativity or something like that can arise so just that question of like Behavior being like too rigid in an environment versus like actually being able to change being able to come up with new things unexpected things even things that haven't been tested so you would expect that like why would you do that that would probably bring higher surprise uh even more uncertainty so that just those kinds of questions balancing those things yeah the the there's there's a ton to to say there

[Music]

um well what one aspect is in expected

free energy so in the Criterion used for prospective selection of action there's an Information Gain or epistemic value term

so surprising information informative information is actively sought after because policies that lead to Information Gain are directly chosen so that's one kind of prospective looking answer um that holds up both seeking observations that that Accord with um preferences that's loaded on to pragmatic value and seeking out informative information that's the information game and I and that's um referenced here at the end of the Box

um another piece to it is it's kind of like all these different seesaws epistemic and pragmatic value or underfitting and overfitting active inference isn't saying there is a specific answer to that but it gives us the the canvas that we could then do the balancing within like you can make an agent that only undertakes epistemic action or conversely under only undertakes pragmatic action so it's not that just framing it within active inference gives you an Adaptive agent it's just that this is an expressive enough framework for generative modeling that you can find those special regions or manifolds within State space that might have those kind of like flexible or adaptive properties so it's not that something

works and and does balance informational
and pragmatic value just because it's
using expected free
energy it has to be kind of
tuned to to to sit at that special
place Dr Einstein then anyone
else okay you're
unmuted I never expected these computers
you know still learning curve since I've
hopped in the time
machine
um but um yeah I think last week we we
towards the end of the session um got to
that that line in box 3.3 this could be
understood as the imperative to explain
things as accurately as possible but
also keep options open and avoid to
committing to any specific explanation
unless this is necessary that's the
maximum entropy
principle um I mean it just yeah that's
such a curveball they kind of sneak in
there at the end I'm still trying to
reconcile that because it makes sense um
intuitively in respect to like
information forging behavior um the
necessity
to to calculate surprise um Beyond just
simply minimizing it or you know in in
respect to expected free energy rather
than variational free energy so it's
kind of I guess an episte it's a it's an
epistem ially in interesting question as
far as what is
the conception of of Truth in the active
inference framework I mean it it seems
like it implies that
everything you know any agent has to

Simply view the world and truths in the world through a probabilistic framework and um there can be no absolute you know a priori truth in in this or am I I mean these These are huge huge philosophical questions yeah suffice to say that it's not about what the territory is or does however we may take a probabilistic approach to modeling A system that has deterministic or probabilistic characters and then whether or not we think this or that about this or that and whether it thinks this or that any one of those could be right or wrong or literally whatever so I think it's it's a jumping off point for some larger philosophical questions but certainly it's not at the fundamental layer of the framework to to have an opinion on stances that even with little familiarity we can see one could jump one way or the other um Andrew yeah sorry just to follow up on that um and then my question would be is you know how how would you how do you and maybe I just need to research more but how to reconcile you know that forcing imperative or maximum entropy imperative with the with the hamiltonian you know gradient of least action right is there a I mean is there a statistical means of reconciling reconciling those Concepts yeah great great question the path of behaviors that the agent under

goes is being conceptualized as a path
of least action most likely thing that
the generative model could do so
depending on the generative models setup
the most likely thing for it to do might
be engage in exploratory
behavior um Andrew then
Colin yeah um we had the what is the low
road versus The High Road question
earlier um and I thought I'd give my
stab at
it um nice the low road is the
fundamental question is what's the
nature of
inference and the high road the
fundamental question is what is the
identity that's my stab I have some
um the for follow on questions um I
voted up a couple that I thought are
interesting and were questions I had um
hopefully they raised to the tap top but
that's
all cool that's awesome because
inference could be about anything it
doesn't have to be about identity so we
could do Bas theorem basian statistics
doesn't have to be about a cybernetic
autopoietic
entity and then identity especially if
we're going to be talking about
Persistence of identity and
self-organization in a dissipate of KNE
and all of this it doesn't tell us how
so I think this is an awesome um diet um
Colin then
Pablo I was just stimulated to think
from looking again at the um the
explanation of expected free energy as

uh a balance between uh Information Gain and pragmatic value and I found these these these yeah all these formula were were interesting in that they they all represent a kind of balancing of two things one of which you could prioritize more um but only by reducing reducing the other and anyway it it just seemed it seemed to me that yeah that we're prescribing the right way in which these two things are to be balanced against each other uh when active inference is done and we were kind of saying therefore that this is the way it must happen but uh I was wondering why is it not possible to take a step back and say well my model about how I decide what to do is I could give greater priority to Information Gain and less to pragmatic value and uh is it not a reasonable thing that people could make choices to do do it differently um in what way is forbidden or made Difficult by active inference

Theory does that make yeah these These are great questions um very nice Insight that that there's like two pieces broken out in each of these framings

um don't don't get hit with the equal time fallacy when you actually calculate this for real policies this one might if if depending on how the numbers are this might equal 0.1 and this might equal a

million so policy selection might be driven by one or the other just because there's two terms doesn't mean that it's

5050 so if the C Vector to be more concrete here is very large and we'll see this later in the textbook if the C Vector has a large size then pragmatic value dominates expected free energy if the C value is small relative to uncertainties then Information Gain dominates pragmatic considerations and so again this is like a seesaw and then adaptivity is like the Seesaw being balanced but just the Seesaw itself is not balanced because that's kind of there's no one on it but when you actually build the generative model and parameterize it then you'll start to play around and see oh maybe something needs to change because it's only doing information foraging it's never going for the pragmatic outcome um I was thinking more in a situation where they were they were closely balanced and that would be that there was a fine tuning of moving more more more waiting one more than the other it seemed like it's been prescribed here how you should wait them in that situation uh and should I spend time getting results or should I abandon getting results uh and go and get more information uh sometimes we actually face that choice do I do more research or do I start writing now and both seem both seem attractive so both have the similar um weight as it were in the decision-making context yeah this is a uh articulation but not a waiting and and there are policies and niches where

maybe a given policy has high pragmatic and informational value so just it if it really if you sketched out the generative model of doing research as some have and when to start or stop or move to a different resource then this articulation would help at least kind of reckon with that and then if it turns out that there's two policies that have a similar EXP expected free energy because one of them is really informative and one is pragmatic then they would have similar posterior probabilities of being selected and that agent would potentially switch between those policies um Pablo then sus yes this chapter it make me think about um that we live in in an entropy uh universe but active uh agents are like entropy Fighters that's kind of right yeah anything that's being repeatedly measured at least has to weather the storm even if it's just an inner particle and then things that are active seem to go even beyond that like they reconstitute themselves and they persist even more than you'd expect for just something and being iner and weathering the storm what does that make you think about it's it's fun it's like uh we are like um as far as we know not many of these agents on on on the universe so I wonder why we it's like uh you know entropy all

around us and and we we're trying to
like uh who we want it or not fight
fight against it and it's it's
cool okay yeah that's that's that's
classical High Road that's kind of the
the mystery of persistence or self of
just
perpetuation and then why active
inferences in the middle is we have a
first principles persistence
imperative with an implementable process
Theory coming from the low road and
that's why the active inference
generative model can be used to address
those kinds of situations of persistence
and it also has a tractable way
otherwise it's just a description of the
problem of persistence without a way to
generate any kind
of agent to actually do
it um

Susan then

Kristoff

um so yeah I like the distinction
between um the nature of identity nature
of inference um and that kind of brings
me
to how to um understand and yeah the
probability versus possibility kind of
comes into my frame of reference because
if I'm looking at the high road through
the nature of
identity then that's going to lead to
you my preferences and my um and my
perceived
importances
so how does that change my Markov
blanket how does what change

it
um if I
am if
if if I'm moving between you know what
I'm what I'm interpreting here is that
if I look at the high road as the nature
of identity then that for me has more to
do with the area of possibility or you
know going back to a statement made um I
think last week in terms of action
starts with um an imag imaginative
process or predictive
representation so that's going to that's
going to drive my perceived affordances
right yes what
what it there's a lot of different ways
to say it for something that's simpler
or more sophisticated but yeah what
actions can I select from what are my
affordances such that I'm in the
business of finding out what I am or
confirming what I am and confirming what
I am might be I'm the kind that goes and
finds out new information that's the
information foraging piece so just self-
evidencing doesn't mean just Naval
gazing or just um shutting down sensory
input as people for years have kind of
repeatedly
raised self- evidencing can be broader
than just seeking confirmation of
sensory evidence in a short term right
and and so how would that would that not
you know let's say that I recognize that
you well hey I could be this way too so
I basically um invoked or updated my
beliefs about what my affordances are
would that not

change my Markov
blanket expand
it am I not
thinking depend it's a little it's a
little understated because a markup
blanket in what sense I mean the sensory
apparatus is not going to change so
let's just say someone says I'm not the
kind of person who goes to parties if it
if it were at a party it wouldn't be
me so that's the kind of identity and
what something expects to be but then if
someone has a different internal model
says I'm the kind of person that
sometimes is out of party and sometimes
isn't then they could self evidence in
either of those settings but in either
of those settings it's they would still
have visual input and hands so their
their Markov blanket wouldn't change
or a generative modeler modeling the
person going to the party or not it
doesn't need to entail a
difference but you could make a
situation where the Markov blanket does
change like I'm the kind of person who
does or doesn't use neural link or does
or doesn't take on this extra sensory
technology because that is a more
structural modification of the interface
of the agent with the niche and that has
a more vital relationship with the
blanket and with the object's um sense
of self perhaps so can I ask one more
question does that so does that lead me
to the boundary of my Markov blanket if
I'm
updating my

affordances sketch out the space and see what it looks like and sketch what is the bound what is the bound and what's what's inside and what's out outside because marov blankets very general any base graph is going to have marov blankets a plenty so then it can be helpful to clarify are we talking about like what kinds of spaces okay um that's the modeling part right yeah yeah okay R go for it hey Susan so your question about um the nature of identity and how can the Markoff blanket of an agent shift around um reminds me of some ideas that I was thinking about recently and this is maybe a bit more of a psychological and uh philosophical uh perspective but thinking about maybe two kinds of of um organisms let's say um biological cells and then also people uh humans that are in a relationship so um the question is about what do we consider at any given moment to be me what is it that I uh say or or have in my mind this is what counts as me and I think you can notice uh if you get into very serious relationship uh romantic relationship or or even having a child uh I imagine and I think I experienced this as well that the that boundary between what I think of as myself um begins to extend and begins to

wrap around uh the other person um and
to the extent that we have uh begin to
have the same shared goals and beliefs
that um the extension of the boundary
becomes more

well well defined and more uh robust and
and the other way that I was thinking
about this is in in cells in biological
cells

how um it seems to be

that uh for example in cancer cancer
cells uh are cells that have separated
from the pable organism have become
alienated in in a sense and no longer um
share the same goal or or know what's
the same um have the same

instructions um uh as as the rest of the
cells in the organism have so they see
the rest of their environments as as
external environments whereas uh usually
yourselves even though each of them are
individuals um they kind of are have

this Synergy because they're all
integrated and have the same uh sort of
sort of objective but um when a cancer
cell kind of gets uh alienated and it
starts to have the same uh intentions

and and goals as as uh a cell that's
what is uh not part of an organism just
kind of an individual cell in the wild
would have which is to um find as as
much nutrition and as possible and then
start to spread and then that's when you
have excessive growth of uh of the cell
so uh the connection I make there is

how um biological cells

also have need to be able to figure out
this process of

redefining what do they think think of
uh how do they think of as themselves
and they manage to do
that um and that's how you end up having
a multicellular
organism that is composed of many uh
individual cells that would otherwise
have their
own um notion of of of what they have to
do in order to persist as as
individuals thank you yes
self-organization is described in the
kind of current
here but then there could be composite
self-organization with multiple layers
so thank you um Kristoff and Andrew
quickly and then we'll look at it try to
look at a few questions before we
end uh I just want to make two two very
quick points one you know um Susan when
you were talking about this notion of
identity just somehow the um uh Serenity
Prayer popped into my head you know go
God grant me the serenity to accept the
things I cannot change the the courage
to change the things they can and the
wisdom to know the difference so what
exactly constitutes your identity it's a
very good question and if you start
looking for it um the answer is really
not obvious um and you know we we in in
my view we draw these um these Mark of
blankets around parts of the graph and
we say like there was some you know
statistical independence of um U certain
variables from certain variables and
that's that's a useful distinction but
it's not actually um it's a kind of in

some sense a virtual uh distinction
basically Mark of blanket is a
perspective um and that perspective can
simply be changed from from from one
perspect because there's always some
kind of hierarchical nested structure
and we can adjust it as as needed so the
notion of identity is is very fluid and
a lot of traditions um you know like
like Buddhist traditions for instance um
teach you how to look for your identity
and you will find that there is no such
thing as identity that boundary cannot
be really drawn
um and you know the the goal of of a lot
of these practices to Simply lead it to
the realization that we're all
interconnected at at some incredibly
deep level and if you're if you're
trying to draw that boundary you're
always going to find some some kind of
way in which the boundary doesn't hold
but it's still nevertheless kind of
useful and to go back to the previous
point about this Information Gain versus
epistemic value the way I started seeing
free energy principles that we start
with Bas theorem and which is kind of
like this
kind of non-computable in in the general
case piece of mathematics and then we
use it to almost total pretty much not
almost but PR topologically carve up um
uh that belief updating in all kinds of
different ways to give us a different
perspective on what what's going on but
it's not it's not an
implementation prescription it's a it's

it's a certain kind of myopic View for
the lens of you know probability
distributions and base VI and so on so
forth kind of geometric View on the
world and how it's actually implemented
whether there is a um uh whether there
is some kind of parameter that weighs um
you know um uh information looking for
epistemic Value versus um um the other
term um that's that that's still you
know that's that's kind of hidden by the
by the formalism and that's necessarily
hidden because the formalism kind of
comes up from basically making this
distinction up out of nowhere right
which just we just take the posterior
expand it through base theorem carve it
up however we like it group some terms
and say like well this is this and this
is that and the distinction is in some
sense imposed on something where that
that that doesn't
necessarily uh you know um it's it's not
inherent in the object it's it's it's
it's an imposed point of
view okay thank you yes Andrew and Dr
Einstein very quickly and then let's see
if we can go to a question
yeah I just have a couple um thoughts on
uh some previous points uh one I forget
who it was but the the comment about
entropy and US kind of be being entropy
Fighting Machines I I just had a thought
that we've probably all been embedded in
this um Milo that treats uh second law
of Thermodynamics as like universally in
inevitable and I think there's some good
evidence due to um like Cosmic expansion

and uh some arguments that um
contemporary thinkers are making that uh
actually order might be more likely um
the second law only applies when you
have a closed container uh that you'll
reach kind of heat death and most of our
evidence in terms of the universe is we
are in open system all the way around
and closed containers are actually the
exception not the rule um I'm going to
post a couple links to books that sort
of talk about that uh what I I first
found it from Julian Barbo um who's kind
of a a Oddball physicist um he's got a
book called The Janice point that he
talks about the relevance for that in
terms of time and um also I believe
Alicia Giro has been is her name pops up
in um the Koda our textbook group um the
context changes everything um and uh
she's got some interesting arguments
that kind of touch on that and also
touch on the whole Markoff blanket uh
dialogue we're having here um I believe
you can get her book for free it's
unlocked I'm currently looking at it and
highly recommend it so far um and
finally the whole um related to this
entropy question um there's some uh
debate going around on a new Theory
called assembly Theory um that some
people think is just utter boulder dash
but uh it's got a new interesting take
on uh the the nature of order um and
worth taking a look at um so yeah and oh
one final thing markof blankets um I I
believe and your Markoff blanket changes
if uh you've ever used a hammer before

and um you just hammer in a nail um
you're you're the change between um
where the blanket ends and begins um and
your body there's evidence that your
mind uh incorporates a tool um from a
car to a hammer to anything in between
and spectacles will change your uh
Markoff blank as

well

thank you yes indeed that's why there's
so much discussion of the extended
embedded cognitive processes because we
can both hold up what the thing is and
understand how tools and so on change it
cognitively Dr Einstein quickly and then
we'll look at one question yeah just
real quickly um I posted a link in the
chat to a conversation uh in his hippo
head on converging dialogues and just on
that that notion of embodiment she does
a really thorough overview of some of
the philosophical underpinnings of the
high road as it uh relates to embodiment
in the work of Merleau-Ponty and Haeghe
Andy Clark so I really really recommend
it um even for non-philosophers um it's
it's very accessible

so and she really touches on that the
the need of or yeah the relationship of
the body and en and the environment you
know as an agent for gathering
information and model updating respect
to Andy Clark's work so check that out
guys

cool in our last minutes um first if you
feel like you have more to say or you
have more questions than answers more
answers than

questions I so heavily and heartedly encourage you to modify the Koda please please add the links there that's how we move together as a niche if you add questions if you add links if you improve links you will learn you will help others learn it's really super important and special um there's a ton of questions of course that we didn't get to but but just even highlighting some of the questions in red these are like epic things that people have asked and and again there's a lot of improvable discourse in a lot of these topics so like I know it's a lot to to um look at and discuss and so we're like all just kind of forging through this massive Niche but no one's going to read all of this so we're always just kind of like finding and leaving things um forward from each other um that's part of the fun let's just look ahead to what's going to happen this was our second discussion on chapter 3 we're going to be heading next into chapter four chapter four is where we get to the generative models of active inference chapter one just background two and three were like the roads into town chapter 4 is the town this is active inference chapter five is going to discuss what's happening in that town in the mammalian Neuroscience District just focusing on a few human or mammal type Neuroscience cases where generative models have been deployed and that'll take us basically

to the end of the
year um people can add their different
ideas and and so on but that's kind of
what the first half of the book the
epistemic half of the book ends up
looking like and then the second half is
bringing closer to the modeling and the
Hands-On and so on with the recipe
starting in chapter six but um yeah it's
quite a journey and the low road and the
high road definitely invoke very
different regimes of attention chapter 4
um is going to be longer and with some
more analytical details than the
previous
chapters

so I encourage people to read it lightly
um which may be slowly but there are
some formalisms and so on that are not
vital for understanding in fact they
even suggest like you can like skip over
this chapter which is very funny that
they say that but um we will return next
week I'll be really looking forward to
the the notes and the questions that
people have so thank you everybody see
you next time bye thanks a lot guys
Halloween everyone